

VALLEY, TX, USA

WMO: 722505

Lat: 26.228N

Lon: 97.654W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12904

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.8	4.8	-5.2	2.4	11.8	-2.4	3.1	10.2	14.4	24.6	12.8	22.9	5.4	340	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.1	37.1	25.3	36.3	25.4	35.5	25.3	27.2	31.9	26.9	31.5	26.6	31.3	7.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	28.1	26.1	21.5	28.0	25.7	20.9	27.9	85.6	31.8	84.2	31.5	82.8	30.9	31.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	11.2	10.2	0.1	38.9	2.1	1.4	-1.4	40.0	-2.6	40.8	-3.8	41.6	-5.4	42.7		
			-1.4	28.2	1.8	0.9	-2.7	28.9	-3.7	29.4	-4.7	29.9	-6.0	30.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	23.8	16.1	18.3	20.9	24.2	27.3	29.2	29.7	30.0	27.8	24.6	20.2	16.8
	DBStd	6.14	5.10	5.18	4.41	3.62	2.58	1.53	1.39	1.44	2.04	3.51	4.52	5.06
	HDD10.0	26	11	5	1	0	0	0	0	0	0	0	2	7
	HDD18.3	326	104	61	27	4	0	0	0	0	0	4	33	92
	CDD10.0	5053	199	236	338	427	535	576	611	619	535	453	307	217
	CDD18.3	2312	34	59	106	181	277	326	352	361	285	199	89	44
	CDH23.3	24266	209	376	719	1568	2905	3884	4358	4467	2958	1854	682	285
	CDH26.7	11105	29	91	210	593	1281	1953	2277	2362	1339	734	187	49
Wind	WSAvg	5.1	5.1	5.6	5.8	5.9	5.8	5.2	5.2	4.6	3.8	4.3	4.6	4.9
Precipitation	PrecAvg	702	42	36	22	41	76	74	50	79	146	63	37	39
	PrecMax	1151	116	185	100	264	304	191	220	228	450	161	110	110
	PrecMin	366	1	0	0	0	0	0	0	6	20	4	0	4
	PrecStd	181	36	36	27	55	67	51	53	58	95	43	32	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.0	31.3	33.0	35.2	36.3	37.4	37.6	37.9	37.3	34.4	32.0	29.7
		MCWB	20.2	21.0	22.2	23.8	25.6	25.8	25.1	25.4	24.8	24.8	23.8	21.9
	2%	DB	27.3	29.1	30.7	33.1	34.7	36.1	36.8	37.2	35.2	33.0	30.3	28.0
		MCWB	20.4	21.5	22.1	23.4	25.2	25.6	25.2	25.5	25.3	24.6	22.8	21.7
	5%	DB	26.0	27.6	29.0	31.9	33.7	35.1	36.0	36.3	34.0	32.1	28.9	26.7
		MCWB	19.5	21.0	21.7	23.3	24.8	25.4	25.3	25.5	25.3	24.3	22.3	21.0
	10%	DB	23.9	25.9	27.4	30.2	32.5	34.1	35.0	35.3	32.8	30.9	27.5	24.8
		MCWB	18.8	20.0	20.8	22.7	24.6	25.3	25.4	25.5	25.2	23.9	21.5	20.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.7	23.4	24.1	26.1	27.0	27.7	27.4	27.5	27.5	26.9	25.3	23.9
		MCDB	26.1	28.0	29.6	31.3	32.9	33.1	31.3	32.7	31.2	30.1	29.5	27.2
	2%	WB	21.7	22.7	23.4	25.1	26.4	27.2	26.8	26.9	26.9	26.3	24.3	23.0
		MCDB	25.0	26.6	28.0	30.5	32.0	32.0	31.2	32.2	31.0	29.4	27.9	26.0
	5%	WB	21.0	21.9	22.8	24.4	25.8	26.8	26.5	26.7	26.6	25.7	23.7	22.3
		MCDB	23.9	25.3	26.8	29.3	30.9	31.4	31.1	32.0	30.8	29.4	26.9	25.2
	10%	WB	20.2	21.3	22.1	23.9	25.5	26.4	26.2	26.3	26.2	25.2	22.9	21.4
		MCDB	23.1	24.4	25.5	28.3	30.2	30.8	31.0	31.6	30.4	29.2	26.0	24.1
Mean Daily Temperature Range	5% DB	MDBR	11.4	11.0	10.9	10.9	10.3	10.5	10.8	11.1	9.9	11.0	11.2	11.1
		MCDBR	12.4	11.8	11.4	11.6	10.9	11.3	12.0	12.0	11.3	11.6	11.7	11.9
		MCWBR	6.5	5.2	4.5	4.0	3.2	2.8	2.7	2.7	3.4	4.3	5.3	6.5
	5% WB	MCDBR	10.5	10.0	9.8	10.0	9.5	9.7	10.4	11.0	9.4	9.4	9.9	10.3
		MCWBR	6.2	5.0	4.5	4.5	3.9	3.2	2.9	2.9	3.0	3.4	4.3	5.3
Clear-Sky Solar Irradiance	taub	0.361	0.375	0.398	0.438	0.449	0.426	0.445	0.431	0.438	0.390	0.376	0.364	
	taud	2.468	2.418	2.353	2.250	2.252	2.378	2.334	2.379	2.325	2.457	2.452	2.461	
	Ebn at Noon	888	901	897	865	850	864	847	860	846	875	865	868	
	Edh at Noon	98	110	124	140	140	123	128	122	125	104	98	95	
All-Sky Solar Radiation	RadAvg	3.11	3.74	4.60	5.46	6.03	6.58	6.56	6.27	5.21	4.56	3.50	2.80	
	RadStd	0.34	0.55	0.49	0.46	0.39	0.24	0.44	0.27	0.37	0.42	0.36	0.38	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105

Nomenclature: See separate page